

SPANNING TREES OF $G \times K_2$ BY VERTICAL SUPPORT, PARTIAL DOUBLES, AND THE BIJECTIVE PROGRAM FOR THE HYPERCUBE TREE FORMULA

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ABSTRACT. For a connected graph G , we count the spanning trees of the prism $G \times K_2$ by their *vertical support* — the set S of vertices of G whose vertical edge is used: the count is $\frac{1}{2} c(G) 2^{|S|} F_G(S)$, where $c(G)$ is the number of spanning trees and $F_G(S)$ the number of S -rooted spanning forests of G . Iterating this identity over the n directions of the hypercube $Q_n = Q_{n-1} \times K_2$ telescopes, through Pascal’s rule, to Stanley’s product formula $c(Q_n) = 2^{2^n - n - 1} \prod_k k^{\binom{n}{k}}$, giving an elementary inductive derivation of that formula and, more importantly, reducing the problem of a fully bijective proof — open since [2, Example 5.6.10], and still open in bijective form after Bernardi’s combinatorial proofs [1] — to a single per-level bijective step. We then prove the *class-level law*: refining by the multiset U of level-projected horizontal edges, the trees in each class (S, U) number exactly $d(S, U) 2^{|S|-1}$, where $d(S, U)$ counts the splittings of U into a spanning tree and an S -rooted forest of G — and each splitting occurs among the trees of its class as an explicit *polarized* representative. This proves every counting statement in the program down to its finest stratum; in the rooted setting we prove an all-spins-zero normal form theorem and its colored-forest generalization, which, combined with the explicit spin-flip involutions furnished by Bernardi’s inductive proof of spin independence, assemble into an explicit algorithmic bijection between rooted spanning trees of Q_n and tuples of colors in $\prod_k [2k]^{\binom{n}{k}}$ — a Prüfer code for the hypercube, and with it a bijective proof of Stanley’s formula. The construction is implemented and machine-verified (exhaustively for $n \leq 3$; sampled at $n = 4$); its one recursive ingredient, the spin normalization, remains to be made closed-form. We also record an exactly verified spin-uniformity phenomenon (Bernardi’s independence holds exactly for rooted-anywhere trees and provably fails for root-fixed ones) that constrains the shape of any such map.

1. BACKGROUND

Stanley’s *Enumerative Combinatorics* vol. 2 derives

$$(1) \quad c(Q_n) = 2^{2^n - n - 1} \prod_{k=1}^n k^{\binom{n}{k}}$$

from the Laplacian eigenvalues of Q_n and remarks (1st ed., p. 62, closing Example 5.6.10) that “a direct combinatorial proof of this formula is not known.” Bernardi [1] gave two combinatorial proofs — one via an independence property of edge-direction “spins” in uniformly random rooted forests, one via killing involutions — and the second edition of [2] (2024, Notes to Ch. 5) updates the remark: a *bijective* proof is not known. This paper organizes a route to one.

Throughout, G is a connected graph on M vertices (for (1), $G = Q_{n-1}$ and $M = 2^{n-1}$), and $G \times K_2$ is the prism over G : two copies $G \times \{0\}$, $G \times \{1\}$ (*horizontal* edges) plus a *vertical*

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edge $\{(u, 0), (u, 1)\}$ for each $u \in V(G)$. For a spanning tree T of $G \times K_2$, let $S(T) \subseteq V(G)$ be its vertical support. For $S \subseteq V(G)$, let $F_G(S)$ denote the number of spanning forests of G with exactly $|S|$ components, each containing one vertex of S ; by the all-minors Matrix-Tree theorem $F_G(S) = \det L_G[\bar{S}]$, the minor of the Laplacian with the rows and columns of S removed.

2. THE SUPPORT-PRODUCT THEOREM

Theorem 2.1. *For every nonempty $S \subseteq V(G)$, the number of spanning trees T of $G \times K_2$ with $S(T) = S$ is*

$$N_G(S) = \frac{1}{2} c(G) 2^{|S|} F_G(S).$$

Proof. Weight the vertical edge at u by an indeterminate z_u and every horizontal edge by 1, and let $t(z) = \sum_T \prod_{u \in S(T)} z_u$ be the weighted spanning-tree sum. With vertices ordered level by level, the weighted Laplacian is $\begin{pmatrix} L_G + Z & -Z \\ -Z & L_G + Z \end{pmatrix}$, $Z = \text{diag}(z_u)$, which the orthogonal level-swap basis $\frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}$ block-diagonalizes as $L_G \oplus (L_G + 2Z)$. For a weighted graph on N vertices the product of the nonzero Laplacian eigenvalues is N times the tree sum, and the kernel lies in the L_G block; with $N = 2M$,

$$t(z) = \frac{1}{2M} \left(\prod_{\lambda \neq 0} \lambda(L_G) \right) \det(L_G + 2Z) = \frac{c(G)}{2} \det(L_G + 2Z).$$

Expanding by principal minors in the diagonal matrix $2Z$ and applying the all-minors Matrix-Tree theorem, $\det(L_G + 2Z) = \sum_R 2^{|R|} F_G(R) \prod_{u \in R} z_u$. Both sides are multilinear in the z_u ; compare coefficients of $\prod_{u \in S} z_u$. $\square$

Remark 2.2. The identity is elementary and product-graph Matrix-Tree computations of this kind appear in Martin–Reiner [3] and Bernardi [1, §4]; we make no novelty claim for Theorem 2.1 itself. Its role here is architectural: it is the exact statement whose bijective realization would prove (1) bijectively (Corollary 2.3 and Problem 7.1). We first found it as an experimental identity: it was confirmed class-by-class for Q_2, Q_3, Q_4 (3+15+255 supports; the Q_4 case over all 679 477 248 tree–root pairs) before the proof was written down.

Corollary 2.3. *Iterating Theorem 2.1 yields (1).*

Proof. Summing the theorem over all S (the $S = \emptyset$ term vanishes) and using $\sum_R z^{|R|} F_G(R) = \det(zI + L_G)$ at $z = 2$:

$$c(G \times K_2) = \frac{1}{2} c(G) \det(2I + L_G).$$

For $G = Q_{n-1}$, whose Laplacian eigenvalues are $2k$ with multiplicity $\binom{n-1}{k}$, $\det(2I + L) = \prod_{k=0}^{n-1} (2k+2)^{\binom{n-1}{k}}$. Writing $R_n = 2^n c(Q_n)$, the recursion reads $R_n = R_{n-1} \prod_{j=1}^n (2j)^{\binom{n-1}{j-1}}$, and Pascal’s rule gives $R_n = \prod_{j=1}^n (2j)^{\binom{n}{j}}$ by induction, which is (1). $\square$

Of course, evaluating $\det(2I + L_{Q_{n-1}})$ spectrally re-uses linear algebra; combinatorially, $\det(2I + L_G) = \sum_S 2^{|S|} F_G(S)$ counts *pairs* (an S -rooted spanning forest, a $\{0, 1\}$ -coloring of its roots), so the recursion has the set-level form

$$(2) \quad \{\text{spanning trees of } G \times K_2\} \times \{0, 1\} \longleftrightarrow \{\text{spanning trees of } G\} \times \{\text{root-2-colored rooted spanning forests of } G\},$$

and a bijective proof of (1) needs exactly a constructive form of (2), level by level.

3. BIJECTIVE FRAGMENTS

3.1. The full-support slice.

Proposition 3.1. *A spanning tree of $G \times K_2$ containing every vertical edge is the same data as a spanning tree of G together with a level in $\{0, 1\}$ for each of its $M - 1$ edges: contract the vertical edges to project, and lift each base tree edge to its recorded level to invert. Hence $N_G(V(G)) = c(G) 2^{M-1}$, the $S = V(G)$ case of Theorem 2.1.*

3.2. Partial doubles.

Definition 3.2. For $S \subseteq V(G)$, the *partial double* $G^{(S)}$ is the graph obtained from G by splitting each $u \notin S$ into u_0, u_1 : an edge (u, v) of G becomes two parallel edges when $u, v \in S$; the two edges $(u, v_0), (u, v_1)$ when $u \in S, v \notin S$; and $(u_0, v_0), (u_1, v_1)$ when $u, v \notin S$.

Contracting the vertical edges of a tree with support S is a bijection onto the spanning trees of $G^{(S)}$, so Theorem 2.1 is equivalent to

$$(3) \quad c(G^{(S)}) = \frac{1}{2} c(G) 2^{|S|} F_G(S).$$

For $S = V(G)$ this is Proposition 3.1 (all edges doubled); for $S = \emptyset$ both sides vanish on connectivity grounds. Splitting one vertex at a time interpolates between $G^{(V)}$ and $G^{(S)}$ and is a natural inductive route to a bijection, though the ratio $F_G(S)/F_G(S \cup u)$ is not local, so the induction must carry more structure than a single count.

3.3. What the spins do and do not give. Bernardi's independence theorem [1, Thm. 1] concerns uniformly random rooted forests conditioned on their projection; transported to spanning trees it says the *spins* (for each $u \in S$, the level of the endpoint of the vertical edge at u nearer the root) should be free bits. We verified computationally: for the ensemble of (tree, root) pairs of Q_3 , conditioned on the direction and the support S , the joint distribution of the spins is *exactly* uniform on $\{0, 1\}^S$ (all 45 conditioned classes, zero discrepancy), and remains consistent with uniform on samples of 2×10^6 (Q_4) and 10^6 (Q_5) uniform trees. For trees with a *fixed* root the uniformity provably fails (e.g. $\chi^2 = 330.7$ on 65 dof already for Q_3). Thus the factor $2^{|S|}$ of Theorem 2.1 is realized by free bits only in the rooted-anywhere ensemble; any bijection must fix the root only after the bits are extracted. This matches the shape of (2), which is stated for a 2-fold cover of the tree set rather than for rooted trees.

4. THE CLASS-LEVEL LAW

Fix S and refine further. Deleting (rather than contracting) the rungs of a tree T' with support S leaves horizontal forests in the two levels; projecting both to G gives an edge multiset U with multiplicities in $\{1, 2\}$: write D for the doubled edges (used in both levels), M for the singletons ($m = |M|$, each carrying a free bit: which level), and $N(S, U)$ for the number of trees with data (S, U) . Call a pair (T, F) a *decomposition* of (S, U) if T is a spanning tree of G , F is an S -rooted spanning forest of G , and $T \uplus F = U$ as multisets; let $d(S, U)$ be their number. Note that a decomposition determines its class: $D = T \cap F$ and $M = T \triangle F$.

Theorem 4.1 (class-level law). *For every connected G , every nonempty $S \subseteq V(G)$, and every multiset U ,*

$$N(S, U) = d(S, U) \cdot 2^{|S|-1}.$$

Proof. Introduce a variable w_e for each edge e of G and z_u for each $u \in V(G)$; in $G \times K_2$ weight both copies of e by w_e and the rung at u by z_u . As in Theorem 2.1, the weighted Laplacian block-diagonalizes as $L_w \oplus (L_w + 2Z)$, where L_w is the weighted Laplacian of (G, w) and $Z = \text{diag}(z_u)$, and the weighted tree sum of the prism, $t(w, z) = \sum_{T'} \prod_e w_e^{\#\text{copies of } e \text{ in } T'} \prod_{\text{rungs } u \in T'} z_u$, equals $\frac{1}{2} \tau(w) \det(L_w + 2Z)$ with $\tau(w)$ the weighted spanning-tree sum of G . Expanding the determinant by principal minors in $2Z$ and applying the weighted all-minors Matrix-Tree theorem,

$$t(w, z) = \frac{1}{2} \tau(w) \sum_{R \subseteq V} 2^{|R|} \varphi_R(w) \prod_{u \in R} z_u,$$

where $\varphi_R(w)$ is the weighted R -rooted spanning-forest sum. Both sides are multilinear in the z_u ; the coefficient of $\prod_{u \in S} z_u$ gives

$$\sum_{T': S(T')=S} \prod_e w_e^{\#\text{copies}} = 2^{|S|-1} \tau(w) \varphi_S(w).$$

Now extract the coefficient of $w^U := \prod_{e \in D} w_e^2 \prod_{e \in M} w_e$. On the left this is $N(S, U)$. On the right, τ and φ_S are multilinear in the w_e (a tree or forest uses each edge at most once), so the coefficient of w^U in the product counts exactly the pairs (T, F) with $T \uplus F = U$, i.e. $d(S, U)$. $\square$

Setting $w \equiv 1$ recovers Theorem 2.1, so the class-level law is the finest member of the family. It was found the other way around: exhaustive computation on Q_3 and Q_4 (all $132 + 2\,459\,160$ classes, from all $42\,467\,328$ spanning trees of Q_4) revealed the pattern $N = d \cdot 2^{m-2s}$ with s equal to the cycle rank of the support of U , and the identity $m - 2\text{rank} = |S| - 1$ (an exercise with the edge budgets) collapsed it to the statement above, after which the proof was forced.

Proposition 4.2 (polarization). *Decompositions are themselves lifts: if (T, F) decomposes (S, U) , then placing $T \cap M$ on level 0 and $F \cap M$ on level 1 (with D on both) yields a spanning tree of $G \times K_2$ with data (S, U) . Hence $d(S, U)$ counts the polarized trees in their class — those whose level-0 graph is a spanning tree of G and whose level-1 graph is an S -rooted forest.*

Proof. Level 0 carries $D \cup (T \cap M) = T$, connected and acyclic; level 1 carries $D \cup (F \cap M) = F$, a forest with $|S|$ components, one per vertex of S . Each rung $u \in S$ joins the single level-0 component to the level-1 component containing u , and those components are pairwise distinct, so the component graph is a star — a tree — and the lift is a spanning tree. Conversely a tree whose level-0 graph is spanning connected and whose level-1 graph is S -rooted reads off a decomposition. $\square$

Proposition 4.3 (explicit bijection when the support is a tree). *If the support of U is a spanning tree of G (equivalently $m = |S| - 1$), then $d \leq 1$; when $d = 1$, every one of the 2^m level assignments yields a spanning tree, and the identity map on the $m = |S| - 1$ free bits, with the polarized lift as base point, realizes Theorem 4.1 bijectively.*

Proof sketch. When $\text{supp } U$ is a spanning tree, T is forced to be that tree and $F = D$, so $d = 1$ iff D is S -rooted. For validity of every assignment, induct: contract a leaf edge of the support lying in D (harmless on both sides), and otherwise a leaf edge $e = (u, \ell) \in M$ has its leaf ℓ isolated in D , hence $\ell \in S$; whichever level e is assigned, the fiber of ℓ attaches to the rest by e and the rung at ℓ , and deleting the fiber reduces to the same situation with $(S \setminus \ell, M \setminus e)$. Both choices of the bit extend every valid smaller assignment, giving the doubling and the explicit bijection. $\square$

5. THE ROOTED NORMAL FORM

The missing map materializes in the *rooted* setting, which the spin experiments of Section 4 already singled out. For a spanning tree L of $G \times K_2$ with a root vertex r and a rung $u \in S(L)$, define the *spin* $\sigma(u) \in \{0, 1\}$ as the level of the endpoint of the rung nearer to r in L .

Theorem 5.1 (all-spins-zero normal form). *A rooted tree (L, r) of class (S, U) has $\sigma \equiv 0$ if and only if L is polarized (Proposition 4.2) and r lies at level 0.*

Proof. The component graph B of L is a tree, *bipartite between levels* (every rung joins a level-0 component to a level-1 component). Root B at the component of r ; then $\sigma(u)$ is the level of the parent-side component of the edge u . If $\sigma \equiv 0$, every parent is a level-0 component; by bipartiteness a level-0 component cannot have a level-0 parent, so no level-0 component has a parent at all: the level-0 side is the single root component, whence H_0 is a spanning tree, and every level-1 component is a leaf with exactly one rung, i.e. exactly one vertex of S : H_1 is S -rooted, and L is polarized with r at level 0. The converse is immediate. $\square$

Counting check: the polarized trees of a class number $d(S, U)$, each with $|V(G)|$ level-0 roots, and $\sigma \equiv 0$ is one of $2^{|S|}$ spin patterns among $2|V(G)| \cdot N(S, U)$ rooted trees; by Theorem 4.1 the two counts agree, so Theorem 5.1 is consistent with — and, given spin-class equinumerosity, equivalent to — the class-level law.

Spin-class equinumerosity within each class is exactly Bernardi's independence theorem [1, Thm. 1], whose inductive proof is constructive: the atomic move toggles the level of the single attaching edge of a pendant fiber (his case of a vertex with no incoming projected arc), and the general spin flip is a composition of such moves organized by his induction. Composing that normalization with Theorem 5.1 and Proposition 4.2 yields:

Theorem 5.2 (rooted straightening). *For every connected G there is a bijection, explicit modulo the inductive unwinding of [1, Thm. 1],*

$$\{(L, r) : L \text{ spanning tree of } G \times K_2, r \text{ a root}\} \longleftrightarrow \{(T, v)\} \times \{(F, \sigma)\},$$

where (T, v) is a rooted spanning tree of G and (F, σ) an S -rooted spanning forest with a $\{0, 1\}$ -color on each root, summed over S : i.e. the set form $2|V| c(G \times K_2) = |V| c(G) \cdot \sum_S 2^{|S|} F_G(S)$ holds by explicit correspondence, class by class.

We verified the constituent claims exhaustively over all 108 384 rooted spanning trees of the prisms over nine base graphs (paths, cycles, cliques, near-cliques; `qpark/scripts/rooted_spins.py`).

6. THE HYPERCUBE PRÜFER CODE

The pieces assemble. Write $\mathcal{F}_z(G)$ for the set of z -colored rooted spanning forests of G (each component carries a root with a color from $[z]$), so $|\mathcal{F}_z(G)| = \det(zI + L_G)$.

Theorem 6.1 (colored hierarchy, constructive). *For every connected G and $z \geq 1$ there is an explicit bijection*

$$\mathcal{F}_z(G \times K_2) \longleftrightarrow \mathcal{F}_z(G) \times \mathcal{F}_{z+2}(G),$$

obtained by composing (i) spin normalization — for each rung of a forest, the spin is the level of its endpoint nearer the root of its component, and Bernardi's induction [1, proof of Thm. 1], specialized to the Cartesian product, furnishes an explicit involution flipping any one spin and nothing else — with (ii) the colored star normal form: in an all-spins-zero forest every rung-containing component collapses, by the bipartite argument of Theorem 5.1 applied per component, to a level-0 rooted piece with level-1 leaves; level 0 then reads off as

an element of $\mathcal{F}_z(G)$ and the level-1 pieces as an element of $\mathcal{F}_{z+2}(G)$, a piece being either own-rooted (old color $< z$) or rung-attached (root at its attachment vertex, color z plus its recorded spin — the two new colors are the two spin values).

Theorem 6.2 (Prüfer code for the hypercube). *Iterating Theorem 6.1 over the n directions of Q_n (the tree factor threading through with $z = 1$ and its forest factor entering the hierarchy at $z = 2$) yields an explicit bijection*

$$\{(T, r) : T \text{ spanning tree of } Q_n, r \text{ a root}\} \longleftrightarrow \prod_{k=1}^n [2k]^{\binom{n}{k}},$$

i.e. every rooted spanning tree of the hypercube is encoded by, and recovered from, one tuple of colors with $\binom{n}{k}$ coordinates of size $2k$. In particular $2^n c(Q_n) = \prod_k (2k)^{\binom{n}{k}}$, which is Stanley’s formula (1), proved bijectively.

The bijection is algorithmic: the spin flip is defined by a terminating recursion (delete-and-restore of pendant fibers and spliced degree-one fibers, following [1]), and every other step is a closed-form local rule. An implementation accompanies this repository (`qpark/scripts/bijection.py`); it has been machine-verified to be a bijection: exhaustively at the per-level statement for six (G, z) combinations (up to 9 800 forests each, image confirmed equal to the full product), exhaustively for $n = 3$ (all 3 072 rooted spanning trees of Q_3 mapped onto the exact box $[2]^3 \times [4]^3 \times [6]$ with perfect round-trips), and on samples for $n = 4$ (200 uniform rooted trees, distinct codes in the box $[2]^4[4]^6[6]^4[8]$ of size 679 477 248, perfect round-trips). Whether a bijection whose one recursive ingredient is unwound rather than closed-form meets every reader’s standard for a “direct” bijective proof is a judgment we leave to the community; making the spin normalization closed-form (Problem 7.1) remains the natural refinement.

7. OPEN PROBLEMS

All counting in the program is now proven, down to the finest stratum. The sole remaining gap between the results above and a fully bijective proof of (1) is a *map*:

Problem 7.1. *Make the spin normalization self-contained and non-recursive for Cartesian prisms: an explicit closed-form bijection between the spin classes of each (S, U) , e.g. by exhibiting the composite of pendant flips directly. This would make Theorem 5.2 fully self-contained.*

Remark 7.2. An earlier draft posed the colored hierarchy as an open problem; it is now Theorem 6.1. The remaining refinement is Problem 7.1 alone.

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